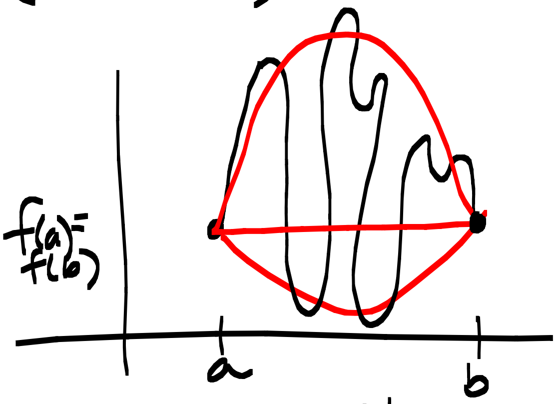
Note Title 2014/04/06

Unit 4.2 The mean value theorem p 284

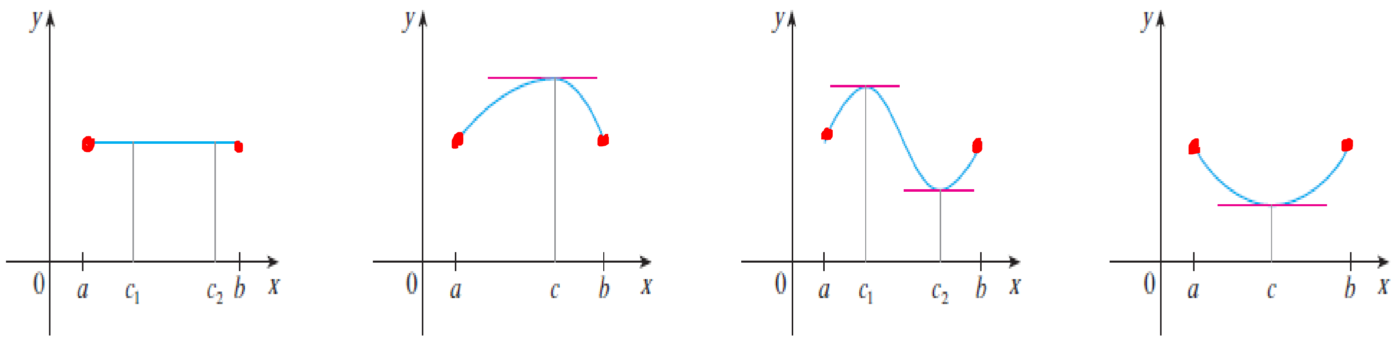
Rolle's Theorem p284 (No proof)

f is a function so that:

1. f continuous on $[a, b]$
2. f differentiable on (a, b)
3. $f(a) = f(b)$



then there exists a number c so that $f'(c) = 0$



(a) (b) (c) (d)

Prove that $f(x) = e^x + x^3$ has just one root.

Let's assume the $f(x)$ has two roots.
So there is an a and b where
$f(a) = f(b) = 0$

Rolle's theorem:

f continuous, differentiable

$f(a) = f(b) = 0$

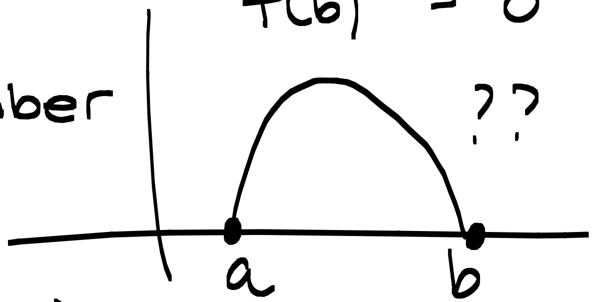
$\Rightarrow$ there exists a number

$c \in (a, b)$ so that

$f'(c) = 0$

$f(a) = 0$

$f(b) = 0$



$\Rightarrow$

Therefore $f'(x) = e^x + 3x^2$ and so

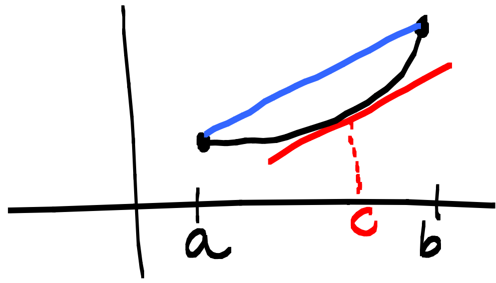
$f'(c) = e^c + 3c^2 = 0$

$\Rightarrow e^c = -3c^2$ which is

never possible $\Rightarrow$ by contradiction

there is only one root.

Mean value theorem p285 (No proof) (MVT)



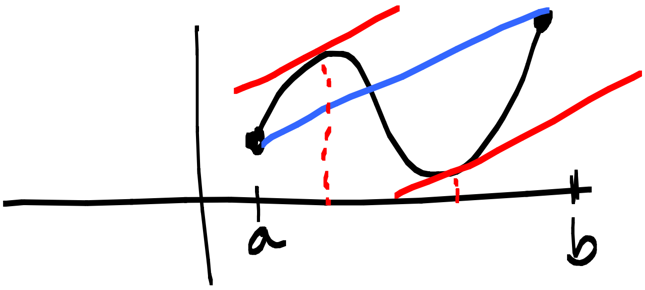
f is a function so that:

① f continuous on $[a, b]$

② f differentiable on (a, b)

then there exists a number $c \in (a,b)$ so that $f'(c) = \dfrac{f(b)-f(a)}{b-a}$ or

$$f(b) - f(a) = f'(c)(b-a)$$



Verify the two conditions required by the MVT (mean value theorem) and then find a suitable number c guaranteed to exist by the MVT

if $P(x) = x^2 + 5x$ on $[0, 2]$

1. P is a polynomial $\Rightarrow$ Continuous on $[0, 2]$

2. $p'(x) = 2x + 5$ is a polynomial so p is differentiable on $(0,2)$

MVT states there exists a number $c \in (0,2)$ so that

$$p'(c) = \frac{p(b) - p(a)}{b - a}$$

$$p'(c) = \frac{p(2) - p(0)}{2 - 0} = \frac{14 - 0}{2 - 0} = 7 \quad *$$

but $p'(x) = 2x + 5$

$$\Rightarrow p'(c) = 2c + 5 \quad **$$

From $(*)$ and $(**)$

$$2c + 5 = 7 \Rightarrow c = 1$$